

Decent Coaching Classes
IX CBSE
Mind Your Math - VI — Answer Key

Section A — MCQ

1. (b) II
2. (b) 1
3. (c) $\sqrt{7}$
4. (b) $4ab$
5. (c) 90°
6. (b) 30 cm^2 [Area = $\frac{1}{2} \times \text{base} \times \text{height} = \frac{1}{2} \times 10 \times 6 = 30 \text{ cm}^2$]

Section B — VSA

7. (0, 5)
8. $3x - 12 = 0 \Rightarrow x = 4$
9. $1/4 = 0.25$ and $1/2 = 0.5$; two rational numbers between them : $0.3 (= 3/10)$ and $0.4 (= 2/5)$
10. Number of tails = $40 - 22 = 18$. $P(\text{tail}) = 18/40 = 9/20$

Section C — SA (any one option in each)

11. $(\sqrt{5} + \sqrt{3})(\sqrt{5} - \sqrt{3}) = (\sqrt{5})^2 - (\sqrt{3})^2 = 5 - 3 = 2$
OR
 $97^2 = (100 - 3)^2 = 100^2 - 2 \times 100 \times 3 + 3^2 = 10000 - 600 + 9 = 9409$
12. Distance from centre = $\sqrt{(r^2 - (AB/2)^2)} = \sqrt{(5^2 - 4^2)} = \sqrt{(25 - 16)} = \sqrt{9} = 3 \text{ cm}$
OR
Radius = $\sqrt{(d^2 + (\text{chord}/2)^2)} = \sqrt{(12^2 + 5^2)} = \sqrt{(144 + 25)} = \sqrt{169} = 13 \text{ cm}$
13. $s = (13 + 14 + 15)/2 = 21$. Area = $\sqrt{[s(s-a)(s-b)(s-c)]} = \sqrt{(21 \times 8 \times 7 \times 6)} = \sqrt{7056} = 84 \text{ cm}^2$
OR
Perimeter = $2(l + b) = 2(24 + 18) = 84 \text{ m}$. Area = $l \times b = 24 \times 18 = 432 \text{ sq. m}$

Section D — Case Study

14. Total balls = $5 + 3 + 2 = 10$
(a) $P(\text{red}) = 5/10 = 1/2$
(b) $P(\text{not green}) = (5 + 3)/10 = 8/10 = 4/5$
(c) $P(\text{blue}) = 3/10$. Expected number in 100 trials = $(3/10) \times 100 = 30$ times